Webcam
~30 fps, frame timestamps
Landmarks
MediaPipe: 478 points, blendshapes
Features
EAR (6 pts/eye), eyeBlink scores
Detector
threshold + hysteresis
Segmenter
duration → reject/dot/dash/cmd
Decoder
gaps → letters, words, commands
Output
text, speech
SYNTHETIC DATA
not study results
everything runs in the browser; no camera frame leaves the device